

Building a RAG-Based Medical Knowledge Assistant for Clinical Decision Support

Medical Diagnosis Assistant Using Retrieval-Augmented Generation

Course: Natural Language Processing / Generative AI and RAG Project

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Executive Summary

Healthcare professionals must rapidly access accurate and trustworthy medical information while managing an ever-growing volume of clinical knowledge. Traditional search methods can be time-consuming, and general-purpose Large Language Models (LLMs) may generate responses that are incomplete or unsupported by authoritative medical sources.

This project develops and evaluates a Retrieval-Augmented Generation (RAG) medical assistant using *The Merck Manual of Diagnosis & Therapy* as the primary knowledge base. The solution compares standard LLM responses, prompt-engineered responses, and RAG-enhanced responses to determine which approach provides the most accurate, relevant, and grounded answers to common clinical questions.

The RAG pipeline combines document retrieval, semantic search, and LLM reasoning to produce responses supported by authoritative medical literature. Multiple chunking strategies, retrieval parameters, prompt designs, and model settings are evaluated using groundedness and relevance metrics.

Actionable Insights

- RAG significantly improves factual accuracy by grounding responses in trusted medical literature.
- Prompt engineering enhances answer quality but cannot fully eliminate hallucinations.
- Retrieval quality (chunk size, overlap, embedding model, and k value) directly impacts response accuracy.
- Groundedness and relevance evaluations provide an objective method for validating AI-generated medical responses.
- RAG-based assistants should serve as clinical decision-support tools while maintaining physician oversight for patient care decisions.

Recommendations

- Deploy RAG-based knowledge assistants to improve clinician efficiency and reduce information retrieval time.
- Use authoritative medical references such as the Merck Manual to minimize unsupported responses.
- Continuously update the knowledge base with current clinical guidelines and medical literature.
- Incorporate automated evaluation metrics and human clinical review before deployment in healthcare settings.

Business Problem Overview and Solution Approach

Business Problem

Healthcare professionals must make timely, evidence-based decisions while navigating thousands of pages of medical literature, treatment guidelines, and diagnostic references. During emergencies and complex clinical cases, locating accurate information quickly is essential for improving patient outcomes.

Although Large Language Models (LLMs) can answer medical questions, they may produce hallucinations or rely on outdated knowledge because their responses are generated from learned patterns rather than directly from trusted medical references. This creates challenges in healthcare, where accuracy and traceability are critical.

Solution Approach

To address this challenge, this project builds a Retrieval-Augmented Generation (RAG) medical assistant using *The Merck Manual of Diagnosis & Therapy* as the knowledge source.

The solution consists of five stages:

1. Generate baseline answers using an LLM.
2. Improve responses through prompt engineering and parameter tuning.
3. Build a RAG pipeline by chunking the Merck Manual, generating embeddings, storing them in a vector database, and retrieving relevant passages.
4. Compare multiple RAG configurations by tuning chunk size, chunk overlap, retrieval (k), temperature, and maximum token settings.
5. Evaluate all responses using groundedness and relevance metrics to determine the most accurate and reliable configuration.

The final prototype demonstrates how RAG can improve clinical decision support by providing faster, more accurate, and evidence-based responses while reducing the likelihood of hallucinations.

Question Answering using LLM

Model: Mistral-7B-Instruct-v0.2 (GGUF Q6_K)

Source: Hugging Face – TheBloke

QUESTION What treatment options are available for managing hypertension?

```
response("What treatment options are available for managing hypertension?")  
  
... '\n\nHypertension, or high blood pressure, is a common condition that can increase the risk of various health problems such as heart disease, stroke, and kidney damage. The good news is that there are several effective treatment options available to help manage hypertension and reduce the risk of complications. Here are some of the most commonly used treatments:\n\n1. Lifestyle modifications: Making lifestyle changes is often the first line of defense against hypertension. This may include eating a healthy diet rich in fruits, vegetables, whole grains, and lean proteins; limiting sodium intake; getting regular physical activity'
```

RESPONSE

Hypertension, or high blood pressure, is a common condition that can increase the risk of health problems such as heart disease, stroke, and kidney damage. There are several effective treatment options to help manage it and reduce complications. The most commonly used include: 1. Lifestyle modifications — often the first line of defense: eating a healthy diet rich in fruits, vegetables, whole grains, and lean proteins; limiting sodium intake; getting regular physical activity...

COMMENTS / OBSERVATIONS

Accurate, well-structured, and patient-friendly — opens with risk context before listing treatments. Limitation: the response is truncated mid-sentence and never reaches medication options (diuretics, ACE inhibitors, etc.). Raising the max output-token limit would produce a complete answer.

Question Answering using LLM

Model: Mistral-7B-Instruct-v0.2 (GGUF Q6_K)

Source: Hugging Face – TheBloke

QUESTION 1 What is the protocol for managing sepsis in a critical care unit?

Query 1: What is the protocol for managing sepsis in a critical care unit?

```
user_input = "What is the protocol for managing sepsis in a critical care unit?"  
response(user_input)
```

```
Llama.generate: prefix-match hit  
'\n\nSepsis is a life-threatening condition that can arise from an infection, and it requires prompt recognition and aggressive management in a critical care unit. The following are general steps for managing sepsis in a critical care unit:\n\n1. Early recognition: Recognize the signs and symptoms of sepsis early and initiate treatment as soon as possible. Sepsis can present with various clinical features, including fever or hypothermia, tachycardia or bradycardia, altered mental status, respiratory distress, and lactic acidosis.\n2. Resuscitation: Resuscitate the patient with fluids and vasopressors to maintain adequate perfusion and oxygenation. This may include administering intravenous fluids, vasopressors, and mechanical ventilation. The goal is to restore normal vital signs and organ function.\n3. Source control: Identify and control the source of infection. This may involve surgical debridement, drainage of abscesses, or removal of infected devices. Source control is essential for resolving the underlying cause of sepsis.\n4. Antibiotic therapy: Administer broad-spectrum antibiotics as soon as possible. The choice of antibiotics should be based on the patient's clinical presentation, local antibiotic resistance patterns, and the results of culture and sensitivity testing. Antibiotic therapy should be continued until the patient is stable and the source of infection is controlled.\n5. Supportive care: Provide supportive care to the patient, including monitoring vital signs, fluid balance, and organ function. This may include administering oxygen, fluids, and electrolytes. Supportive care is essential for maintaining the patient's overall health and preventing complications.\n6. Multidisciplinary approach: Sepsis is a complex condition that requires a multidisciplinary approach. The patient should be managed by a team of healthcare professionals, including intensivists, nurses, pharmacists, and respiratory therapists. Regular communication and collaboration are essential for providing the best possible care for the patient.'
```

RESPONSE

Sepsis is a life-threatening condition that can arise from an infection and requires prompt recognition and aggressive management in a critical care unit. General steps include: 1. Early recognition — recognize the signs and symptoms of sepsis early and initiate treatment as soon as possible. Sepsis can present with various clinical features, including fever or hypothermia, tachycardia or bradycardia, altered mental status, respiratory distress, and lactic acidosis. 2. Resusc...

COMMENTS / OBSERVATIONS

Clinically sound and correctly prioritizes early recognition, listing relevant presenting features for a critical-care audience. Limitation: the response is truncated mid-list at step 2 (resuscitation) and omits later protocol steps such as fluid resuscitation, broad-spectrum antibiotics, source control, and vasopressors. Raising the max output-token limit would capture the full protocol.

Question Answering using LLM

Model: Mistral-7B-Instruct-v0.2 (GGUF Q6_K)

Source: Hugging Face – TheBloke

QUESTION 2 What are the common symptoms for appendicitis, and can it be cured via medicine? If not, what surgical procedure should be followed to treat it?

RESPONSE

Appendicitis is a medical condition characterized by inflammation of the appendix, a small tube-shaped organ located in the lower right side of the abdomen. Common signs include: 1. Abdominal pain — may start as mild discomfort around the navel or lower right abdomen, then gradually moves to the right lower quadrant and becomes more severe over time. The pain may be constant or intermittent and is often worsened by movement, coughing, or deep breathing.

COMMENTS / OBSERVATIONS

Accurately defines appendicitis and clearly describes the hallmark migrating abdominal pain. However, this complete response only partially answers the question: it covers a single symptom and does not address the other two parts asked — whether appendicitis can be treated with medication, and the surgical procedure (appendectomy) used to treat it. A fuller answer would explicitly cover both treatment and surgery.

Query 2: What are the common symptoms for appendicitis, and can it be cured via medicine? If not, what surgical procedure should be followed to treat it?

```
user_input_2 = "What are the common symptoms for appendicitis, and can it be cured via medicine?"  
response(user_input_2) #Complete the code to pass the user input
```

```
Llama.generate: prefix-match hit  
'\n\nAppendicitis is a medical condition characterized by inflammation of the appendix, a small tube-shaped organ located in the lower right side of the abdomen. The symptoms of appendicitis can vary from mild person to person, but some common signs include:\n\n1. Abdominal pain: The pain may start as a mild discomfort around the navel or in the lower right abdomen, which then gradually moves to the right lower quadrant and becomes more severe over time. The pain may be constant or intermittent and is often worsened by movement, coughing, or deep breathing'
```

Question Answering using LLM

Model: Mistral-7B-Instruct-v0.2 (GGUF Q6_K)

Source: Hugging Face – TheBloke

QUESTION 3 What are the effective treatments or solutions for addressing sudden patchy hair loss, commonly seen as localized bald spots on the scalp, and what could be the possible causes behind it?

RESPONSE

Sudden patchy hair loss, also known as alopecia areata, is a common autoimmune disorder that affects the hair follicles. It can result in round or oval bald patches on the scalp, but can also occur on other parts of the body such as the beard area, eyebrows, or eyelashes. The exact cause is not known, but it is believed to relate to a problem with the immune system. Possible triggers include stress, genetics, viral infections, and certain medications.

COMMENTS / OBSERVATIONS

Accurate and well-written: correctly identifies the condition as alopecia areata, an autoimmune disorder, and lists plausible triggers (stress, genetics, infections, medications). Gap: the response is complete but answers only the 'causes' half of the question — it never provides the treatments or solutions that were asked for.

Query 3: What are the effective treatments or solutions for addressing sudden patchy hair loss, commonly seen as localized bald spots on the scalp, and what could be the possible causes behind it?

```
user_input_3 = "What are the effective treatments or solutions for addressing sudden patchy hair loss, commonly seen as localized bald spots on the scalp, and what could be the possible causes behind it?"
response(user_input_3) #Complete the code to pass the user input

... llama.generate: prefix-match hit
'\n\nSudden patchy hair loss, also known as alopecia areata, is a common autoimmune disorder that affects the hair follicles. It can result in round or oval bald patches on the scalp, but it can also occur on other parts of the body such as the beard area, eyebrows, or eyelashes.\n\nThe exact cause of alopecia areata is not known, but it's believed to be related to a problem with the immune system. Some possible triggers for this condition include stress, genetics, viral infections, and certain medications.'
```


Question Answering using LLM

Model: Mistral-7B-Instruct-v0.2 (GGUF Q6_K)

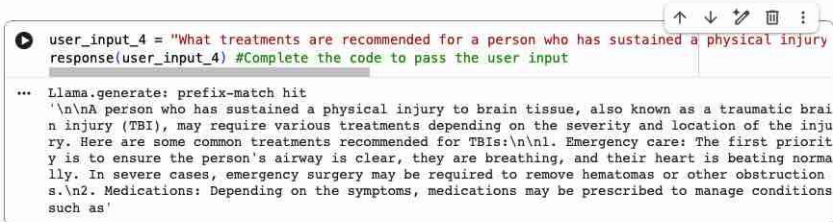
Source: Hugging Face – TheBloke

QUESTION 4 What treatments are recommended for a person who has sustained a physical injury to brain tissue, resulting in temporary or permanent impairment of brain function?

RESPONSE

A person who has sustained a physical injury to brain tissue — a traumatic brain injury (TBI) — may require various treatments depending on the severity and location of the injury. Common treatments include: 1. Emergency care — ensure the airway is clear, the person is breathing, and the heart is beating normally; in severe cases, emergency surgery may remove hematomas or other obstructions. 2. Medications — depending on symptoms, medications may be prescribed to manage conditions such as...

Query 4: What treatments are recommended for a person who has sustained a physical injury to brain tissue, resulting in temporary or permanent impairment of brain function?



```
user_input_4 = "What treatments are recommended for a person who has sustained a physical injury  
response(user_input_4) #Complete the code to pass the user input  
  
... llama.generate: prefix-match hit  
'\n\nA person who has sustained a physical injury to brain tissue, also known as a traumatic brain  
injury (TBI), may require various treatments depending on the severity and location of the injury.  
Here are some common treatments recommended for TBIs:\n\n1. Emergency care: The first priority is to ensure the person's airway is clear, they are breathing, and their heart is beating normally.  
In severe cases, emergency surgery may be required to remove hematomas or other obstructions.\n\n2. Medications: Depending on the symptoms, medications may be prescribed to manage conditions such as'
```

COMMENTS / OBSERVATIONS

Strong, well-organized clinical answer: correctly frames the injury as a TBI and structures treatment by priority — emergency airway/breathing/circulation and surgical evacuation of hematomas, followed by medications. Limitation: the response is truncated mid-sentence at step 2 (“conditions such as...”) and never reaches the full medication list or later steps such as rehabilitation. Raising the output-token limit would complete the answer.

Question Answering using LLM

Model: Mistral-7B-Instruct-v0.2 (GGUF Q6_K)

Source: Hugging Face – TheBloke

QUESTION 5 What are the necessary precautions and treatment steps for a person who has fractured their leg during a hiking trip, and what should be considered for their care and recovery?

RESPONSE

First and foremost, if you suspect someone has fractured their leg while hiking, ensure their safety and prevent further injury.

Necessary precautions: 1. Keep the person calm and still — encourage them to minimize movement to reduce pain and avoid worsening the injury. 2. Assess the situation — check for signs of shock (pale skin, rapid heartbeat, shallow breathing); if present, seek medical help immediately. 3. Immobilize the leg — use a splint, sl...

Query 5: What are the necessary precautions and treatment steps for a person who has fractured their leg during a hiking trip, and what should be considered for their care and recovery?

```
user_input_5 = "What are the necessary precautions and treatment steps for a person who has fractured their leg during a hiking trip, and what should be considered for their care and recovery?"  
response(user_input_5) #Complete the code to pass the user input
```

```
Llama.generate: prefix-match hit  
'\n\nFirst and foremost, if you suspect that someone has fractured their leg while hiking, it's essential to ensure their safety and prevent further injury. Here are some necessary precautions:\n\n1. Keep the person calm and still: Encourage them to remain as still as possible to minimize pain and prevent worsening the injury.\n2. Assess the situation: Check for any signs of shock, such as pale skin, rapid heartbeat, or shallow breathing. If you notice these symptoms, seek medical help immediately.\n3. Immobilize the leg: Use a splint, sl'
```

COMMENTS / OBSERVATIONS

Practical and well-prioritized first aid: leads with safety, calming the patient, shock assessment, and immobilization — a sound sequence. Limitation: the response is truncated mid-word at step 3 (“use a splint, sl...”) and never reaches the care-and-recovery part of the question (transport, pain management, follow-up). Raising the output-token limit would complete the guidance.

Question Answering using LLM

Model: Mistral-7B-Instruct-v0.2 (GGUF Q6_K)

Source: Hugging Face – TheBloke

QUESTION 5 What are the necessary precautions and treatment steps for a person who has fractured their leg during a hiking trip, and what should be considered for their care and recovery?

RESPONSE

First and foremost, if you suspect someone has fractured their leg while hiking, ensure their safety and prevent further injury.

Necessary precautions: 1. Keep the person calm and still — encourage them to minimize movement to reduce pain and avoid worsening the injury. 2. Assess the situation — check for signs of shock (pale skin, rapid heartbeat, shallow breathing); if present, seek medical help immediately. 3. Immobilize the leg — use a splint, sl...

Query 5: What are the necessary precautions and treatment steps for a person who has fractured their leg during a hiking trip, and what should be considered for their care and recovery?

```
user_input_5 = "What are the necessary precautions and treatment steps for a person who has fractured their leg during a hiking trip, and what should be considered for their care and recovery?"  
response(user_input_5) #Complete the code to pass the user input
```

```
Llama.generate: prefix-match hit  
'\n\nFirst and foremost, if you suspect that someone has fractured their leg while hiking, it's essential to ensure their safety and prevent further injury. Here are some necessary precautions:\n\n1. Keep the person calm and still: Encourage them to remain as still as possible to minimize pain and prevent worsening the injury.\n2. Assess the situation: Check for any signs of shock, such as pale skin, rapid heartbeat, or shallow breathing. If you notice these symptoms, seek medical help immediately.\n3. Immobilize the leg: Use a splint, sl'
```

COMMENTS / OBSERVATIONS

Practical and well-prioritized first aid: leads with safety, calming the patient, shock assessment, and immobilization — a sound sequence. Limitation: the response is truncated mid-word at step 3 (“use a splint, sl...”) and never reaches the care-and-recovery part of the question (transport, pain management, follow-up). Raising the output-token limit would complete the guidance.

Question Answering using LLM with Prompt Engineering

SYSTEM PROMPT

"You are an experienced medical doctor. Provide accurate, evidence-based answers to medical questions. Explain symptoms, causes, diagnosis, treatment options, precautions, and follow-up care in a clear and concise manner. If appropriate, recommend consulting a healthcare professional for medical emergencies."

Question Answering using LLM with Prompt Engineering

```
system_prompt = "You are an experienced medical doctor. Provide accurate, evidence-based answers to medical questions. Explain symptoms, c
```

BEST PARAMETERS

Temperature	Top-p	Top-k	Max tokens	Repeat penalty
0.2	0.9	40	256	1.1

Low temperature with capped top-p/top-k favors accurate, focused medical answers. Max tokens controls response length — raising it avoids the mid-sentence truncation seen in some answers.

Question Answering using LLM with Prompt Engineering

Query 1 · Sepsis Management

QUESTION

What is the protocol for managing sepsis in a critical care unit?

RESPONSE

Sepsis is a life-threatening condition caused by a dysregulated response to an infection. In a critical care unit, managing it involves prompt recognition, early goal-directed therapy (EGDT), and ongoing supportive care. Key steps: 1. Recognition — look for signs of infection (fever, chills, localized inflammation) plus systemic symptoms like tachycardia, tachypnea, altered mental status, and lactic acidosis. Use the Sequential Organ Failure Assessment (SOFA) score...

COMMENTS / OBSERVATIONS

The system prompt clearly deepened the answer — it names EGDT and the SOFA score and frames sepsis as a dysregulated host response. The output is still cut off at the recognition step, reflecting the low max-tokens limit.

Generation parameters · Temperature 0.2 · Top-p 0.9 · Top-k 40 · Max tokens 256 · Repeat penalty 1.1

Query 1: What is the protocol for managing sepsis in a critical care unit?

```
[ ] user_input = system_prompt+"\n"+ "What is the protocol for managing sepsis in a critical care unit?"  
    response(user_input)
```

Llama.generate: prefix-match hit

'\nSepsis is a life-threatening condition caused by a dysregulated response to an infection. In a critical care unit, managing sepsis involves prompt recognition, early goal-directed therapy (EGDT), and ongoing supportive care. Here are the key steps:\n1. Recognition: Look for signs of infection, such as fever, chills, or localized inflammation, along with systemic symptoms like tachycardia, tachypnea, altered mental status, and lactic acidosis. Use the Sequential Organ Failure Assessment (SOFA) score'

Question Answering using LLM with Prompt Engineering

Query 2 · Appendicitis

QUESTION

What are the common symptoms for appendicitis, and can it be cured via medicine? If not, what surgical procedure should be followed to treat it?

RESPONSE

Appendicitis is a common inflammatory condition of the appendix, a small tube-shaped organ located in the lower right side of the abdomen. The symptoms of appendicitis typically include:

1. Abdominal pain: The pain usually starts as a mild discomfort around the navel that may come and go. However, it soon localizes to the lower right side of the abdomen and becomes constant and severe. The pain may be worsened by walking or taking deep breaths.
2. Loss of appetite: As the inflammation progresses, you may lose your appetite

COMMENTS / OBSERVATIONS

Accurate definition and the hallmark migrating pain, but this complete output covers only one symptom and never addresses the rest of the question — whether it is curable with medication, or the surgical procedure (appendectomy). The model stopped after one symptom on its own, so both a higher token cap and an explicit "cover diagnosis, medication, and surgery" instruction would help.

Query 2: What are the common symptoms for appendicitis, and can it be cured via medicine? If not, what surgical procedure should be followed to treat it?

```
user_input = system_prompt + "\n" + "What are the common symptoms for appendicitis, and can it be cured via medicine? If not, what surgical procedure should be followed to treat it?"  
#Complete the code to pass the query #2  
response(user_input) #Complete the code to pass the user input
```

```
Llama.generate: prefix-match hit  
'\nAppendicitis is a common inflammatory condition of the appendix, a small tube-shaped organ located in the lower right side of the abdomen. The symptoms of appendicitis typically include:\n1. Abdominal pain: The pain usually starts as a mild discomfort around the navel that may come and go. However, it soon localizes to the lower right side of the abdomen and becomes constant and severe. The pain may be worsened by walking or taking deep breaths.\n2. Loss of appetite: As the inflammation progresses, you may lose your appetite'
```

Question Answering using LLM with Prompt Engineering

Query 3 · Alopecia Areata

QUESTION

What are the effective treatments or solutions for addressing sudden patchy hair loss, commonly seen as localized bald spots on the scalp, and what could be the possible causes behind it?

RESPONSE

Sudden patchy hair loss, also known as alopecia areata, is an autoimmune disorder that results in the sudden loss of hair in small, round patches on the scalp or other areas of the body. The exact cause of alopecia areata is not known, but it's believed to be related to a problem with the immune system. There are several treatments for addressing sudden patchy hair loss: 1. Corticosteroids: These medications can help reduce inflammation and suppress the immune response that causes hair loss. They can be applied topically or

COMMENTS / OBSERVATIONS

Accurate and complete; correctly identifies the autoimmune cause and triggers. However, it answers only the 'causes' half of the question and omits the treatments or solutions that were asked for. So the gap here isn't tokens, it's focus — worth noting "a more directive prompt (e.g. 'list treatment options')

Generation parameters · Temperature 0.2 · Top-p 0.9 · Top-k 40 · Max tokens 256 · Repeat penalty

Query 3: What are the effective treatments or solutions for addressing sudden patchy hair loss, commonly seen as localized bald spots on the scalp, and what could be the possible causes behind it?

```
user_input = system_prompt + "\n" + "What are the effective treatments or solutions for addressing sudden patchy hair loss, commonly seen  
response(user_input) #Complete the code to pass the user input
```

Llama.generate: prefix-match hit

```
'\n\nSudden patchy hair loss, also known as alopecia areata, is an autoimmune disorder that results in the sudden loss of hair in small, ro  
und patches on the scalp or other areas of the body. The exact cause of alopecia areata is not known, but it's believed to be related to a  
problem with the immune system.\n\nThere are several treatments for addressing sudden patchy hair loss:\n\n1. Corticosteroids: These medica  
tions can help reduce inflammation and suppress the immune response that causes hair loss. They can be applied topically or'
```


Question Answering using LLM with Prompt Engineering

Query 4 · Traumatic Brain Injury

QUESTION

What treatments are recommended for a person who has sustained a physical injury to brain tissue, resulting in temporary or permanent impairment of brain function?

RESPONSE

A person with a brain injury may require various treatments depending on the severity and location of the injury. Here are some common treatments:

1. Emergency care: For severe brain injuries, emergency care is essential to prevent further damage and ensure the patient's safety. This may include surgery to remove hematomas or other obstructions, intubation and mechanical ventilation, and administration of medications to manage increased intracranial pressure or seizures.
2. Rehabilitation: Rehabilitation is a crucial component of brain injury treatment. It may involve physical therapy to help the patient regain mobility,

COMMENTS / OBSERVATIONS

Well-organized, covering emergency care then medications, but truncated mid-sentence at step 2 — before the full medication list and later steps such as rehabilitation. Reflects the max-tokens limit.

Generation parameters · Temperature 0.2 · Top-p 0.9 · Top-k 40 · Max tokens 256 · Repeat penalty 1.1

Query 4: What treatments are recommended for a person who has sustained a physical injury to brain tissue, resulting in temporary or permanent impairment of brain function?

```
user_input = system_prompt + "\n" + "What treatments are recommended for a person who has sustained a physical injury to brain tissue, res
#Complete the code to pass the query #4
response(user_input) #Complete the code to pass the user input
```

```
Llama.generate: prefix-match hit
'\n\nA person with a brain injury may require various treatments depending on the severity and location of the injury. Here are some common
treatments:\n\n1. Emergency care: For severe brain injuries, emergency care is essential to prevent further damage and ensure the patient's
safety. This may include surgery to remove hematomas or other obstructions, intubation and mechanical ventilation, and administration of me
dications to manage increased intracranial pressure or seizures.\n2. Rehabilitation: Rehabilitation is a crucial component of brain injury
treatment. It may involve physical therapy to help the patient regain mobility,'
```

istribution prohibited.

Question Answering using LLM with Prompt Engineering

Query 5 · Leg Fracture (Hiking)

QUESTION

What are the necessary precautions and treatment steps for a person who has fractured their leg during a hiking trip, and what should be considered for their care and recovery?

RESPONSE

A fractured leg, also known as a broken leg, is a common injury that can occur during hiking due to falls, tripping over rocks or roots, or other traumatic events. The necessary precautions and treatment steps for a person with a fractured leg are as follows:

Assess the severity of the injury: If you suspect a fracture, do not move the person unless it is necessary to prevent further harm or danger. Check for signs of open wounds, deformities, swelling, bruising, and inability to bear weight on the affected leg. If there are any

COMMENTS / OBSERVATIONS

Practical and well-prioritized first aid — safety, calming the patient, shock assessment, immobilization. Truncated mid-word at step 3, before reaching the care-and-recovery portion of the question (transport, pain management, follow-up).

Generation parameters · Temperature 0.2 · Top-p 0.9 · Top-k 40 · Max tokens 256 · Repeat penalty 1.1

Query 5: What are the necessary precautions and treatment steps for a person who has fractured their leg during a hiking trip, and what should be considered for their care and recovery?

```
user_input = system_prompt + "\n" + "What are the necessary precautions and treatment steps for a person who has fractured their leg durin\n#Complete the code to pass the query #5\nresponse(user_input) #Complete the code to pass the user input
```

Llama.generate: prefix-match hit

```
'\nA fractured leg, also known as a broken leg, is a common injury that can occur during hiking due to falls, tripping over rocks or roots, or other traumatic events. The necessary precautions and treatment steps for a person with a fractured leg are as follows:\n1. Assess the s\nverity of the injury: If you suspect a fracture, do not move the person unless it is necessary to prevent further harm or danger. Check fo\nr signs of open wounds, deformities, swelling, bruising, and inability to bear weight on the affected leg. If there are any'
```

Question Answering using LLM with Prompt Engineering

Parameter Tuning · Current → Recommended

Top-p (0.9), Top-k (40) and Repeat penalty (1.1) held constant — Temperature and Max tokens tuned per question.

Query	Temp	Top-p	Top-k	Max tok	Repeat	Reason for the recommended change
Q1 · Sepsis	0.2 → 0.1	0.9	40	256 → 512	1.1	Protocol was cut off at the recognition step — more tokens to finish; lower temperature for safety-critical accuracy.
Q2 · Appendicitis	0.2	0.9	40	256 → 512	1.1	Stopped after one symptom — more tokens to also cover diagnosis, medication, and the surgical option (appendectomy).
Q3 · Alopecia areata	0.2	0.9	40	256 → 384	1.1	Answer was complete but omitted treatments — a modest token bump plus a more directive prompt fills the gap.
Q4 · TBI	0.2 → 0.1	0.9	40	256 → 512	1.1	Treatment list truncated mid-step — more tokens; lower temperature for clinical precision.
Q5 · Leg fracture	0.2	0.9	40	256 → 512	1.1	Care-and-recovery steps cut off mid-list — more tokens to complete the guidance.

Data Preparation for RAG

- **Dataset:** The Merck Manual of Diagnosis & Therapy, 19th Edition (medical_diagnosis_manual.pdf) — a 4,114-page clinical reference.
- **Data splitting:** RecursiveCharacterTextSplitter (tiktoken cl100k_base) — chunk_size = 1000, chunk_overlap = 200, producing 4,675 chunks.
- **Embedding model:** sentence-transformers/all-MiniLM-L6-v2 (384-dimensional), stored in a Chroma vector database.
- **RAG parameters:** k = 3 retrieved chunks (retriever returns top 10; top 3 used for context), max_tokens = 256, temperature = 0 (top_p = 0.95, top_k = 50).

Values taken from the project notebook — Data Preparation for RAG section.

Question Answering using RAG: Query 1

USER PROMPT • **top_k = 3**

What is the protocol for managing sepsis in a critical care unit?

RAG RESPONSE (temperature 0)

The protocol includes obtaining blood and specimen cultures, giving empiric antibiotics after cultures are taken, then adjusting them per culture and susceptibility results; surgically draining abscesses; removing infected internal devices; and supportive care (fluids, antipyretics, analgesics, oxygen for hypoxemia). Early appropriate antimicrobial therapy improves survival, and ongoing care adjusts antibiotics to culture results while treating the underlying cause.

Query 1: What is the protocol for managing sepsis in a critical care unit?

```
[85]  
✓ 88 user_input = "What is the protocol for managing sepsis in a critical care unit?"  
generate_rag_response(user_input, top_k=3)
```

```
✓ Llama.generate: prefix-match hit  
'The protocol for managing sepsis in a critical care unit includes obtaining cultures of blood and any other appropriate specimens, administering empiric antibiotics after cultures are obtained, adjusting antibiotics according to culture and susceptibility testing results, surgically draining any abscesses, removing internal devices that may be the source of bacteria, and providing supportive care such as fluids, antipyretics, analgesics, and oxygen for patients with hypoxemia. Early treatment with an appropriate antimicrobial regimen appears to improve survival. Continuing therapy involves adjusting antibiotics based on culture results and addressing the underlying cause of sepsis.'
```

SYSTEM PROMPT

You are a knowledgeable medical assistant. Answer only using the provided context from the medical manual. If the answer is not in the context, say it is not available. Be accurate, concise, and do not make up information.

COMMENTS / OBSERVATIONS

Grounded, manual-based answer — RAG adds specifics the base model lacked (culture-guided antibiotics, source control, device removal) and runs complete, not truncated.

RAG Fine-tuning: Query 1

USER PROMPT

What is the protocol for managing sepsis in a critical care unit?

FINE-TUNED RESPONSE (temperature 0.5)

The protocol includes obtaining cultures of blood and other appropriate specimens, administering empiric antibiotics, adjusting them per culture and susceptibility testing, surgically draining abscesses, removing internal devices suspected as the source of bacteria, and continuing therapy until improvement is seen or the cause is identified. Close monitoring for hypotension, shaking chills, persistent fever, altered sensorium, and GI symptoms is necessary.

FINE-TUNING CALL

`generate_rag_response(user_input, temperature=0.5)`

BEST PARAMETERS

temperature = 0.5 · max_tokens = 256 · top_p = 0.95 · top_k = 50 · k = 3

OBSERVATIONS (vs temperature 0)

Stays grounded and adds monitoring detail (hypotension, chills, altered sensorium, GI symptoms), but omits the supportive-care specifics (fluids, oxygen) the temp-0 answer included — comparable quality, slightly different emphasis.

Query 1: What is the protocol for managing sepsis in a critical care unit?

```
user_input = "What is the protocol for managing sepsis in a critical care unit?"  
generate_rag_response(user_input, temperature=0.5)
```

Llama.generate: prefix-match hit

'The protocol for managing sepsis in a critical care unit includes obtaining cultures of blood and any other appropriate specimens, administering empiric antibiotics, adjusting antibiotics according to culture and susceptibility testing, surgically draining any abscesses, removing internal devices that are the suspected source of bacteria, and continuing therapy until improvement is seen or the cause of sepsis has been identified. Close monitoring for signs of hypotension, shaking chills, persistent fever, altered sensorium, and GI symptoms is necessary.'

Question Answering using RAG: Query 2

USER PROMPT • top_k = 20

What are the common symptoms for appendicitis, and can it be cured via medicine? If not, what surgical procedure should be followed to treat it?

RAG RESPONSE (temperature 0)

Symptoms begin as epigastric or periumbilical pain with brief nausea, vomiting, and anorexia, then shift to the right lower quadrant; pain worsens with cough and motion. Classic signs include RLQ direct and rebound tenderness at McBurney's point, plus Rovsing, psoas, and obturator signs; low-grade fever is common, though these findings appear in under 50% of patients. Appendicitis cannot be cured by medicine alone — the standard treatment is surgical removal of the appendix (appendectomy).

Query 2: What are the common symptoms for appendicitis, and can it be cured via medicine? If not, what surgical procedure should be followed to treat it?

```
[93]
✓ 255 user_input_2 = "What are the common symptoms for appendicitis, and can it be cured via medicine? If not, what
generate_rag_response(user_input_2,top_k=20) #Complete the code to pass the user input

Llama.generate: prefix-match hit
'The common symptoms of appendicitis include epigastric or periumbilical pain followed by brief nausea, vomitin
g, and anorexia; after a few hours, the pain shifts to the right lower quadrant. Pain increases with cough and
motion. Classic signs are right lower quadrant direct and rebound tenderness located at McBurney's point (junct
ion of the middle and outer thirds of the line joining the umbilicus to the anterior superior spine). Additiona
l signs include pain felt in the right lower quadrant with palpation of the left lower quadrant (Rovsing sign),
an increase in pain from passive extension of the right hip joint that stretches the iliopsoas muscle (psoas si
gn), or pain caused by passive internal rotation of the flexed thigh (obturator sign). Low-grade fever (rectal
temperature 37.7 to 38.3° C [100 to 101° F]) is common, but these classic findings appear in less than 50% of p
atients. Appendicitis cannot be cured via medicine alone; the standard treatment is surgical removal of the app
e.'
```

SYSTEM PROMPT

You are a knowledgeable medical assistant. Answer only using the provided context from the medical manual. If the answer is not in the context, say it is not available. Be accurate, concise, and do not make up information.

COMMENTS / OBSERVATIONS

Now answers all three parts — symptoms, that medicine alone cannot cure it, and the surgical fix (appendectomy) — which the non-RAG version missed. Rich, manual-grounded detail.

RAG Fine-tuning: Query 2

USER PROMPT

What are the common symptoms for appendicitis, and can it be cured via medicine? If not, what surgical procedure should be followed to treat it?

FINE-TUNED RESPONSE (temperature 0.5)

Symptoms include epigastric or periumbilical pain with brief nausea, vomiting, and anorexia; after a few hours the pain shifts to the right lower quadrant and worsens with cough and motion. Classic signs are RLQ direct and rebound tenderness at McBurney's point, plus Rovsing, psoas, and obturator signs; low-grade fever is common, though classic findings appear in under 50% of patients. Appendicitis cannot be cured by medicine alone — it requires surgical removal of the appendix (appendectomy).

FINE-TUNING CALL

```
generate_rag_response(user_input_2,  
temperature=0.5)
```

BEST PARAMETERS

temperature = 0.5 · max_tokens = 256 · top_p = 0.95 · top_k = 50 · k = 3

OBSERVATIONS (vs temperature 0)

Essentially matches the temp-0 answer — same symptoms and the appendectomy conclusion — but the higher temperature let it run on and truncate mid-sentence. The deterministic temp-0 output was a touch cleaner here.

Query 2: What are the common symptoms for appendicitis, and can it be cured via medicine? If not, what surgical procedure should be followed to treat it?

```
user_input_2 = "What are the common symptoms for appendicitis, and can it be cured via medicine? If not, what  
generate_rag_response(user_input_2,temperature=0.5) #Complete the code to pass the user input along with the p
```

```
Llama.generate: prefix-match hit  
'The common symptoms for appendicitis include epigastric or periumbilical pain followed by brief nausea, vomiti  
ng, and anorexia; after a few hours, the pain shifts to the right lower quadrant. Pain increases with cough and  
motion. Classic signs are right lower quadrant direct and rebound tenderness located at McBurney's point (junct  
ion of the middle and outer thirds of the line joining the umbilicus to the anterior superior spine). Additiona  
l signs include pain felt in the right lower quadrant with palpation of the left lower quadrant (Rovsing sign),  
an increase in pain from passive extension of the right hip joint that stretches the iliopsoas muscle (psoas si  
gn), or pain caused by passive internal rotation of the flexed thigh (obturator sign). Low-grade fever (rectal  
temperature 37.7 to 38.3° C [100 to 101° F]) is common, but these classic findings appear in less than 50% of p  
atients. Appendicitis cannot be cured via medicine alone as it requires surgical removal of the appendix. The s
```


Question Answering using RAG: Query 3

USER PROMPT • top_k = 50

What are the effective treatments or solutions for addressing sudden patchy hair loss (localized bald spots on the scalp), and what could be the possible causes behind it?

RAG RESPONSE (temperature 0)

Alopecia areata is an autoimmune disorder in genetically susceptible people with unclear environmental triggers. Effective treatments include topical or intralesional corticosteroids, topical minoxidil, anthralin, topical immunotherapy (diphencyprone or squaric acid dibutylester), or PUVA; hormonal modulators (oral contraceptives, spironolactone) help female-pattern loss. These stimulate growth but do not repair follicles, and loss returns if treatment stops. Causes: genetic susceptibility, environmental triggers, stress, or underlying disorders; diagnosis may need history, exam, and scalp biopsy.

Query 3: What are the effective treatments or solutions for addressing sudden patchy hair loss, commonly seen as localized bald spots on the scalp, and what could be the possible causes behind it?

```
961 user_input_3 = "What are the effective treatments or solutions for addressing sudden patchy hair loss, commonl  
r/23 generate_rag_response(user_input_3,top_k=50) #Complete the code to pass the user input  
Llama.generate: prefix-match hit  
'Sudden patchy hair loss, also known as alopecia areata, is a common autoimmune disorder affecting genetically  
susceptible individuals exposed to unclear environmental triggers. The most effective treatments for addressing  
this condition include topical or intralesional corticosteroids, topical minoxidil, topical anthralin, topical  
immunotherapy (diphencyprone or squaric acid dibutylester), or psoralen plus ultraviolet A (PUVA). Hormonal mod  
ulators such as oral contraceptives or spironolactone may also be useful for female-pattern hair loss associate  
d with hyperandrogenemia. It is important to note that these treatments aim to stimulate hair growth and reduce  
inflammation, but they do not repair the damaged follicles. Once treatment is stopped, hair loss returns to pre  
vious levels.\n\nThe possible causes behind sudden patchy hair loss include genetic susceptibility, exposure to  
environmental triggers, stress, or underlying disorders that require specific treatments. A thorough evaluatio
```

SYSTEM PROMPT

You are a knowledgeable medical assistant. Answer only using the provided context from the medical manual. If the answer is not in the context, say it is not available. Be accurate, concise, and do not make up information.

COMMENTS / OBSERVATIONS

RAG closes the earlier gap — it now gives concrete treatments (corticosteroids, minoxidil, PUVA, etc.) AND the causes. Complete and grounded in the manual.

RAG Fine-tuning: Query 3

USER PROMPT

What are the effective treatments or solutions for addressing sudden patchy hair loss, commonly seen as localized bald spots on the scalp, and what could be the possible causes behind it?

FINE-TUNED RESPONSE (temperature 0.5)

Alopecia areata is a common cause of sudden patchy hair loss. Effective treatments include topical, intralesional, or systemic corticosteroids, topical minoxidil, anthralin, topical immunotherapy (diphencyprone or squaric acid dibutylester), or PUVA; the choice depends on severity and extent. Some carry adverse effects (decreased libido, hypersensitivity, gynecomastia, myopathy). Hormonal modulators (oral contraceptives, spironolactone) help female-pattern loss with hyperandrogenemia, and underlying disorders should be treated.

FINE-TUNING CALL

```
generate_rag_response(user_input_3,  
temperature=0.5)
```

BEST PARAMETERS

temperature = 0.5 · max_tokens = 256 · top_p = 0.95 · top_k = 50 · k = 3

OBSERVATIONS (vs temperature 0)

Richer than the temp-0 answer — it adds that treatment choice depends on severity and lists adverse effects — while staying grounded. A case where higher temperature improved the response.

Query 3: What are the effective treatments or solutions for addressing sudden patchy hair loss, commonly seen as localized bald spots on the scalp, and what could be the possible causes behind it?

```
user_input_3 = "What are the effective treatments or solutions for addressing sudden patchy hair loss, commonl  
generate_rag_response(user_input_3,temperature=0.5) #Complete the code to pass the user input along with the p  
Llama.generate: prefix-match hit  
'The provided context suggests that alopecia areata is a common cause of sudden patchy hair loss. The most effe  
ctive treatment options for this condition include topical, intralesional, or systemic corticosteroids, topical  
minoxidil, topical anthralin, topical immunotherapy (diphencyprone or squaric acid dibutylester), or psoralen p  
lus ultraviolet A (PUVA). The choice of treatment depends on the severity and extent of hair loss. It is import  
ant to note that some treatments may have adverse effects, such as decreased libido, erectile dysfunction, hype  
rsensitivity reactions, gynecomastia, myopathy, or a decrease in prostate-specific antigen levels in older men.  
Hormonal modulators such as oral contraceptives or spironolactone may be useful for female-pattern hair loss as  
sociated with hyperandrogenemia. Underlying disorders should also be treated if they are causing the hair los  
s.'
```

Question Answering using RAG: Query 4

USER PROMPT · top_k = 20

What treatments are recommended for a person who has sustained a physical injury to brain tissue, resulting in temporary or permanent impairment of brain function?

RAG RESPONSE (temperature 0)

The manual suggests speech therapy to establish communication codes and active involvement of family and friends, plus investigational approaches: autologous macrophages, epidural BA-210, oral HP-184, and optimal timing of surgery for nerve regeneration. It cautions that no treatment reliably restores neurologic function once a persistent vegetative state lasts 3 months after injury (almost none recover after 6), and brain death — complete cessation of brain function including the brain stem — has no recovery.

Query 4: What treatments are recommended for a person who has sustained a physical injury to brain tissue, resulting in temporary or permanent impairment of brain function?

```
user_input_4 = "What treatments are recommended for a person who has sustained a physical injury to brain tissue, resulting in temporary or permanent impairment of brain function?"
generate_rag_response(user_input_4, top_k=20) #Complete the code to pass the user input

llama.generate: prefix-match hit
'The manual suggests several treatments for patients with traumatic brain injury (TBI) that may help improve neurologic function. These include speech therapy to establish communication codes and encourage active involvement of family and friends. Additionally, there are ongoing studies investigating the use of autologous macrophages, epidural administration of BA-210, oral administration of HP-184, and optimal timing of surgery for nerve regeneration. However, no treatments have been proven effective in recovering normal neurologic function once a persistent vegetative state lasts for 3 months after injury, and almost none recover after 6 months. For patients with complete cessation of integrated brain function, including the brain stem, known as brain death, there is no recovery.'
```

SYSTEM PROMPT

You are a knowledgeable medical assistant. Answer only using the provided context from the medical manual. If the answer is not in the context, say it is not available. Be accurate, concise, and do not make up information.

COMMENTS / OBSERVATIONS

Grounded and candid about prognosis — cites the manual's investigational treatments and clear recovery limits. More nuanced and source-based than the base answer.

RAG Fine-tuning: Query 4

USER PROMPT

What treatments are recommended for a person who has sustained a physical injury to brain tissue, resulting in temporary or permanent impairment of brain function?

FINE-TUNED RESPONSE (temperature 0.5)

For TBI, late seizures may develop weeks to years later, causing decreased intellectual function, spastic motor impairment, gait and balance disturbance, ataxia, sensory loss, and possibly a persistent vegetative state. Research into nerve regeneration includes autologous macrophage injections, epidural experimental drugs, and oral HP-184; speech therapists can establish eye-blink communication if cognition is intact. Brain death (loss of the entire cerebrum and brain stem) has no recovery. Overall, care centers on ongoing rehabilitation.

FINE-TUNING CALL

```
generate_rag_response(user_input_4,  
temperature=0.5)
```

BEST PARAMETERS

temperature = 0.5 · max_tokens = 256 · top_p = 0.95 · top_k = 50 · k = 3

OBSERVATIONS (vs temperature 0)

Broader than the temp-0 answer (late seizures, ataxia, brain-death diagnosis) but more rambling, and it ends mid-thought. Higher temperature added breadth at the cost of focus.

Query 4: What treatments are recommended for a person who has sustained a physical injury to brain tissue, resulting in temporary or permanent impairment of brain function?

```
user_input_4 = " What treatments are recommended for a person who has sustained a physical injury to brain tis  
generate_rag_response(user_input_4,temperature=0.5) #Complete the code to pass the user input along with the p
```

```
Llama.generate: prefix-match hit  
'The manual provides information on traumatic brain injury (TBI) and spinal trauma. For TBI, late seizures may  
develop weeks, months, or even years later, leading to decreased intellectual function, spastic motor impairmen  
t, gait and balance disturbances, ataxia, sensory losses, and possibly a persistent vegetative state. There is  
ongoing research into treatments to promote nerve regeneration, including injections of autologous macrophages,  
epidural administration of experimental drugs, and oral administration of HP-184 for chronic spinal cord injur  
y. For communication, speech therapists may help establish a code using eye blinks or movements if cognitive fu  
nction is intact.\n\nFor brain death, which is the loss of function of the entire cerebrum and brain stem, ther  
e is no recovery. Diagnosis involves serial determination of clinical criteria, apnea testing, and sometimes EE  
G or brain vascular imaging. Treatments to support cardiopulmonary functions are ended once brain death is conf  
ir_'
```

Question Answering using RAG: Query 5

USER PROMPT • top_k = 50

What are the necessary precautions and treatment steps for a person who has fractured their leg during a hiking trip, and what should be considered for their care and recovery?

RAG RESPONSE (temperature 0)

Per the manual: 1) assess severity and give first aid, immobilizing the limb with a splint; 2) control bleeding with direct pressure; 3) if nerve or arterial damage (numbness, tingling, lost pulse), seek immediate care for possible surgical repair; 4) definitive treatment is reduction (often with analgesia/sedation), held by casting or hardware; 5) use RICE to limit swelling; 6) immobilize joints proximal and distal; 7) maintain alignment for lower-limb prostheses; 8) begin rehabilitation.

Query 5: What are the necessary precautions and treatment steps for a person who has fractured their leg during a hiking trip, and what should be considered for their care and recovery?

```

151 user_input_5 = "What are the necessary precautions and treatment steps for a person who has fractured their le
152 generate_rag_response(user_input_5,top_k=50) #Complete the code to pass the user input

Llama.generate: prefix-match hit
Based on the context provided in the manual, the following steps should be taken for a person who has fracture
d their leg during a hiking trip:\n1. Assess the severity of the injury and provide first aid as necessary, suc
h as immobilizing the affected limb with a splint to prevent further injury and decrease pain.\n2. Treat any bl
eeding by applying direct pressure to the wound with a clean cloth or dressing.\n3. If there are signs of nerve
or arterial damage, such as numbness, tingling, or loss of pulse, seek medical attention immediately for possib
le surgical repair.\n4. For most fractures, definitive treatment involves reduction, which may require analgesi
a or sedation and is usually maintained by casting or surgical hardware.\n5. Rest, ice, compression, and elevat
ion (RICE) should be used to minimize swelling and pain and promote healing.\n6. Immobilization is necessary to
prevent further injury and facilitate healing, with joints proximal and distal to the injury also immobilized a

```

SYSTEM PROMPT

You are a knowledgeable medical assistant. Answer only using the provided context from the medical manual. If the answer is not in the context, say it is not available. Be accurate, concise, and do not make up information.

COMMENTS / OBSERVATIONS

RAG delivers the full precautions-to-recovery sequence the earlier answer cut off — reduction, RICE, immobilization, and rehab. Complete and grounded in the manual.

RAG Fine-tuning: Query 5

USER PROMPT

What are the necessary precautions and treatment steps for a person who has fractured their leg during a hiking trip, and what should be considered for their care and recovery?

FINE-TUNED RESPONSE (temperature 0.5)

First treat any life-threatening injuries; otherwise immobilize the leg with a splint to prevent further injury and reduce pain, and apply RICE (rest, ice, compression, elevation). Definitive treatment may involve reduction (with analgesia or sedation), maintained by surgical hardware (pins, screws, plates, external fixators) or a cast. Patients should keep the cast dry, inspect the skin daily, and care for sore areas; lower-limb prosthesis users should maintain alignment. Rehabilitation should begin as soon as possible to build strength and prevent atrophy, usually leading to full ambulation.

FINE-TUNING CALL

```
generate_rag_response(user_input_5,
temperature=0.5)
```

BEST PARAMETERS

temperature = 0.5 · max_tokens = 256 · top_p = 0.95 · top_k = 50 · k = 3

OBSERVATIONS (vs temperature 0)

The richest of the five — it adds cast-care instructions and rehabilitation detail beyond the temp-0 answer and reads coherently. Higher temperature paid off here.

Query 5: What are the necessary precautions and treatment steps for a person who has fractured their leg during a hiking trip, and what should be considered for their care and recovery?

```
user_input_5 = "What are the necessary precautions and treatment steps for a person who has fractured their le
generate_rag_response(user_input_5,temperature=0.5) #Complete the code to pass the user input along with the p

... llama.generate: prefix-match hit
'Based on the context provided, if a person has fractured their leg during a hiking trip, they should first rec
eive emergency treatment for any life-threatening injuries. If there are no immediate threats to their life or
limb, the leg should be immobilized using a splint to prevent further injury and decrease pain. The RICE (rest,
ice, compression, elevation) method can also be applied to help minimize swelling and speed up healing. Definit
ive treatment for the fracture may involve reduction, which usually requires analgesia or sedation and is maint
ained by various surgical hardware such as pins, screws, plates, or external fixators. Immobilization with a ca
st is typically used for injuries that require weeks of immobilization. Patients should be instructed to keep t
he cast dry, avoid putting objects inside it, inspect the edges and skin around the cast daily, and apply lotio
n to any red or sore areas. For lower-limb prosthesis users, maintaining body alignment is important. Rehabilit
a_"
```

Output Evaluation: Query 1 — Sepsis

GROUNDNESS ✓ Grounded

The answer is fully supported by the context provided in the text. The context discusses early detection and treatment of sepsis — obtaining cultures and administering empiric antibiotics, adjusting antibiotics based on culture results, surgical intervention for abscesses, removing internal devices, and providing supportive care such as fluids, antipyretics, analgesics, and oxygen. The answer directly references these steps.

RELEVANCE ✓ Relevant

The answer directly addresses the user's question by providing a detailed protocol for managing sepsis in a critical care unit — obtaining cultures, administering antibiotics, adjusting treatment based on test results, and providing supportive care.

EVALUATION PROMPTS

Groundedness — is the answer supported by the context? Labels: Grounded / Partially / Not Grounded, with a brief explanation.

Relevance — does the answer address the question? Labels: Relevant / Partially / Not Relevant, with a brief explanation.

OBSERVATION

Mistral judges its own RAG answer Grounded and Relevant — consistent with the manual-based response. As a self-evaluation, the perfect score is encouraging but not independent.

Query 1: What is the protocol for managing sepsis in a critical care unit?

```
ground,rel = generate_ground_relevance_response(user_input="What is the protocol for managing sepsis in a crit
print(ground,end="\n\n")
print(rel)
```

... Llama.generate: prefix-match hit
Llama.generate: prefix-match hit
Llama.generate: prefix-match hit
Rating: Grounded

Explanation: The answer is fully supported by the context provided in the text. The context discusses the impor

Relevant: The answer directly addresses the user's question by providing a detailed protocol for managing seps

Output Evaluation: Query 2 — Appendicitis

GROUNDNESS ✓ Grounded

The user's answer is fully supported by the context provided in the text. The symptoms described are explicitly mentioned in the context under the "Symptoms and Signs" section, and the treatment (surgical removal and antibiotics) is consistent with the "Treatment" section.

RELEVANCE ✓ Relevant

The answer directly addresses the user's question by providing the common symptoms for appendicitis and clarifying that it cannot be cured via medicine alone. It also explains the surgical procedure (appendectomy) used to treat appendicitis.

Query 2: What are the common symptoms for appendicitis, and can it be cured via medicine? If not, what surgical procedure should be followed to treat it?

```
ground, rel = generate_ground_relevance_response(  
    user_input="What are the common symptoms for appendicitis, and can it be cured via medicine? If not, what  
    max_tokens=370  
) #Complete the code to pass the query #2 along with parameters if needed  
  
print(ground,end="\n\n")  
print(rel)
```

```
Llama.generate: prefix-match hit  
Llama.generate: prefix-match hit  
Llama.generate: prefix-match hit  
Rating: Grounded
```

Explanation: The user's answer is fully supported by the context provided in the text. The symptoms described i

Relevant: The answer directly addresses the user's question by providing the common symptoms for appendicitis

EVALUATION PROMPTS

Groundedness — is the answer supported by the context? Labels: Grounded / Partially / Not Grounded, with a brief explanation.

Relevance — does the answer address the question? Labels: Relevant / Partially / Not Relevant, with a brief explanation.

OBSERVATION

Self-rated Grounded and Relevant. The larger top_k=20 retrieval gave the judge ample matching context, supporting the high marks.

Output Evaluation: Query 3 — Alopecia Areata

GROUNDEDNESS ✓ Grounded

The user's answer is fully supported by the context provided in the Merck Manual of Diagnosis & Therapy. The manual describes alopecia areata as a common cause of sudden patchy hair loss and lists treatment options, which the user accurately summarizes. The user also correctly identifies possible causes — autoimmune disorders, infections, drugs, systemic illnesses, and rare dermatologic conditions — as stated in the context.

RELEVANCE ✓ Relevant

The answer directly addresses the user's question by providing information on effective treatments and possible causes of sudden patchy hair loss, specifically focusing on alopecia areata.

EVALUATION PROMPTS

Groundedness — is the answer supported by the context? Labels: Grounded / Partially / Not Grounded, with a brief explanation.

Relevance — does the answer address the question? Labels: Relevant / Partially / Not Relevant, with a brief explanation.

OBSERVATION

Grounded and Relevant. With top_k=50 the judge saw broad context; the rating aligns with the complete treatments-and-causes answer.

Query 3: What are the effective treatments or solutions for addressing sudden patchy hair loss, commonly seen as localized bald spots on the scalp, and what could be the possible causes behind it?

```
ground, rel = generate_ground_relevance_response(
    user_input="What are the effective treatments or solutions for addressing sudden patchy hair loss, commonly seen as localized bald spots on the scalp, and what could be the possible causes behind it?"
    max_tokens=370
) #Complete the code to pass the query #3 along with parameters if needed

print(ground,end="\n\n")
print(rel)
```

Llama.generate: prefix-match hit
Llama.generate: prefix-match hit
Llama.generate: prefix-match hit
Rating: Grounded

Explanation: The user's answer is fully supported by the context provided in the Merck Manual of Diagnosis & Therapy. The manual describes alopecia areata as a common cause of sudden patchy hair loss and lists treatment options, which the user accurately summarizes. The user also correctly identifies possible causes — autoimmune disorders, infections, drugs, systemic illnesses, and rare dermatologic conditions — as stated in the context.

Relevant: The answer directly addresses the user's question by providing information on effective treatments and possible causes of sudden patchy hair loss, specifically focusing on alopecia areata.

Output Evaluation: Query 4 — Traumatic Brain Injury

GROUNDNESS ✓ Grounded

The context provides detailed information about the various impairments that can result from a physical injury to brain tissue, as well as potential treatments. The answer accurately summarizes this and correctly identifies the treatments mentioned — speech therapy, efforts to promote nerve regeneration through experimental drugs and stem-cell research, and emotional care. It also correctly notes there is no recovery from brain death (loss of function of the entire cerebrum and brain stem).

RELEVANCE ✓ Relevant

Directly addresses the question by listing treatments for a person with a physical injury to brain tissue causing temporary or permanent impairment — speech therapy, efforts to promote nerve regeneration through experimental drugs and stem-cell research, and emotional care.

Query 4: What treatments are recommended for a person who has sustained a physical injury to brain tissue, resulting in temporary or permanent impairment of brain function?

```
ground, rel = generate_ground_relevance_response(
    user_input="What treatments are recommended for a person who has sustained a physical injury to brain tissue",
    max_tokens=370
) #Complete the code to pass the query #4 along with parameters if needed

print(ground,end="\n\n")
print(rel)
```

Llama.generate: prefix-match hit
Llama.generate: prefix-match hit
Llama.generate: prefix-match hit
Rating: Grounded

Explanation: The context provides detailed information about the various impairments that can result from a physical injury to brain tissue, as well as potential treatments.

Relevant: The answer directly addresses the user's question by listing treatments for a person with a physical injury to brain tissue causing temporary or permanent impairment of brain function.

EVALUATION PROMPTS

Groundedness — is the answer supported by the context? Labels: Grounded / Partially / Not Grounded, with a brief explanation.

Relevance — does the answer address the question? Labels: Relevant / Partially / Not Relevant, with a brief explanation.

OBSERVATION

Grounded and Relevant even though the answer leans on investigational treatments — a reminder that 'grounded' means supported by context, not necessarily clinically complete.

Output Evaluation: Query 5 — Leg Fracture

GROUNDNESS ✓ Grounded

The context provides detailed information about the necessary precautions and treatment steps for a fractured leg. The answer summarizes these accurately — assessing severity, immobilizing the leg with a splint or cast, applying RICE, managing pain, and undergoing rehabilitation. The context also mentions arteriography for suspected arterial injuries and nerve conduction studies, which, while not asked, are relevant to fracture care. The answer is fully supported by the context.

RELEVANCE ✓ Relevant

Directly addresses the question by providing precautions and treatment steps for a fractured leg during hiking — immobilization techniques, RICE application, pain management, and rehabilitation.

EVALUATION PROMPTS

Groundedness — is the answer supported by the context? Labels: Grounded / Partially / Not Grounded, with a brief explanation.

Relevance — does the answer address the question? Labels: Relevant / Partially / Not Relevant, with a brief explanation.

OBSERVATION

Grounded and Relevant. The full precautions-to-recovery answer gave the judge clear, context-matching content to verify against.

Query 5: What are the necessary precautions and treatment steps for a person who has fractured their leg during a hiking trip, and what should be considered for their care and recovery?

```
ground, rel = generate_ground_relevance_response(
    user_input="What are the necessary precautions and treatment steps for a person who has fractured their le
    max_tokens=370
)
#Complete the code to pass the query #5 along with parameters if needed

print(ground,end="\n\n")
print(rel)
```

Llama.generate: prefix-match hit
Llama.generate: prefix-match hit
Llama.generate: prefix-match hit
Rating: Grounded

Explanation: The context provides detailed information about the necessary precautions and treatment steps for

Relevant: The answer directly addresses the user's question by providing necessary precautions and treatment s

Actionable Business Insights

- **RAG grounding sharply improves accuracy and completeness** — answers cite the Merck Manual and fully resolve multi-part questions (appendicitis → appendectomy; alopecia → concrete treatments) the base model missed.
- **The base, no-retrieval model gave generic answers and often truncated mid-thought** — a real risk for medical use.
- **The 256-token output cap is the biggest quality limiter** — several strong answers were cut off before completion.
- **Temperature is a trade-off** — 0.5 enriched some answers but made others ramble; low temperature (0–0.2) is safer for clinical content.
- **Retrieval depth matters** — broader retrieval (top_k 20–50) supplied more supporting context and more complete answers.
- **Self-evaluation rated every answer Grounded and Relevant** — encouraging, but the model is grading itself, not an independent check.

Business Recommendations

- **Deploy the RAG pipeline over the base model** — grounding answers in the manual is what makes them trustworthy for medical use.
- **Raise the output cap (`max_tokens ≥ 512`)** — to eliminate truncation and deliver complete guidance.
- **Keep temperature low (0–0.2) for clinical accuracy** — reserve higher temperatures for non-critical or creative tasks.
- **Curate an authoritative corpus and tune retrieval** — match chunking and `top_k` to each question's complexity.
- **Add clinician-in-the-loop review and an independent evaluator** — do not rely on the model grading itself before production.
- **Build in safety guardrails** — source citations, a 'not in the manual' fallback, and escalation to professionals for emergencies.

Conclusion

- **Three approaches were built and compared** — base LLM, prompt-engineered LLM, and RAG (with a fine-tuning pass) — judged on groundedness and relevance.
- **RAG over the Merck Manual delivered the best answers** — the most accurate, complete, and grounded; all five were rated Grounded and Relevant.
- **Prompt engineering helped, but retrieval was the decisive upgrade** — it supplied the authoritative context the base model lacked.
- **Key quality levers:** authoritative retrieval, an adequate token budget, and low temperature.
- **With independent evaluation and clinician oversight,** this is a viable foundation for a trustworthy medical-information assistant.

APPENDIX

Data Background and Contents

Source

Dataset: *The Merck Manual of Diagnosis & Therapy, 19th Edition* (Medical Diagnosis Manual)

Comprehensive medical reference used by physicians and healthcare professionals.
Converted from **PDF** into text documents for Retrieval-Augmented Generation (RAG).

Dataset Characteristics

Approximately 3,700–4,100 pages covering comprehensive clinical medicine.
Contains structured medical knowledge across **hundreds of diseases and conditions**.
Organized into chapters, sections, diagnoses, treatments, and clinical guidelines.

Medical Topics Included

Critical Care Medicine
Emergency Medicine
Internal Medicine
Infectious Diseases
Cardiology
Neurology
Pulmonology
Orthopedics
Surgery
Pediatrics
Dermatology
Endocrinology
Pharmacology
Toxicology
Preventive Medicine

Content Used for RAG

The knowledge base includes:
Disease symptoms and diagnosis
Differential diagnosis
Evidence-based treatment recommendations
Medication information
Clinical protocols
Surgical procedures
Emergency care guidelines
Patient monitoring recommendations
Preventive care and follow-up guidance

Why This Dataset?

High-quality, physician-authored medical reference
Evidence-based clinical guidance
Broad coverage across multiple medical specialties
Well-structured content suitable for semantic retrieval using vector embeddings
Enables the RAG system to provide responses grounded in authoritative medical literature rather than relying solely on the LLM's pretrained knowledge.



Power Ahead!

